🏗️ Interactive Planar Grapher: User Manual

This tool is designed for the interactive creation, routing, and sign-management of planar multigraphs. It uses a three-mode system to ensure a clean workflow for complex graph architectures.

1. System Requirements

To run the application, ensure your environment meets these minimal configurations:

- **Python 3.7+**
- **Libraries:** Install the graph logic engine via terminal:
  `pip install networkx`
- **System Tool:** Install **Graphviz** (graphviz.org) and ensure it is added to your system **PATH**. This is required for the "Auto-Layout" feature.

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2. The Three-Mode Workflow

The program cycles through three distinct modes. Use the **Spacebar** to toggle between them.

🟢 Mode 1: VERTEX (Placement & Movement)

- **Add Node:** Left-click on any empty space.
- **Move Node:** Click and drag a black node to move it. All connected edges will follow dynamically.
- **Delete Node:** Click a node (it turns yellow) and press the **'D'** key.

🔵 Mode 2: EDGE (Connections)

- **Create Edge:** Click the first node (turns yellow), then click the second node.
- **Planarity Guard:** The system will block the connection and show an error if the move violates mathematical planarity.
- **Default Sign:** New edges use the "Default Type" (Blue/+ or Red/-). Press **'T'** to toggle the default before drawing.

🟠 Mode 3: ROUTE / EDIT (Styling & Routing)

- **Select Edge:** Click the orange handle (midpoint) of an edge. The edge will glow orange.
- **Manual Routing:** Drag the orange handle to curve the edge (Quadratic Bézier) around obstacles or other nodes.

- **Toggle Sign:** While an edge is selected (orange), press **'T'** to flip it between **Positive (Blue Solid)** and **Negative (Red Dashed)**.
- **Delete Edge:** While an edge is selected, press **'D'**.

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3. Global Features

- **Auto-Layout (Untangle):** Click the "Auto-Layout" button at any time. The program will use a force-directed algorithm to reposition nodes and clear up visual clutter.
- **Multigraph Support:** You can draw multiple edges between the same two nodes; use the **Route** mode to pull them apart so they are both visible.

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4. Troubleshooting

- **"Graphviz Error":** Ensure Graphviz is installed on the OS (not just the Python library) and that your terminal recognizes the command `dot -V`.
- **"Non-Planar Error":** This occurs when a connection is mathematically impossible to draw without crossing lines, regardless of how you curve them. Try repositioning nodes and attempting the connection again.

Quick Reference Table

Action	**Shortcut**	**Mode Required**
Cycle Modes	`Spacebar`	Any
Delete Selected	`D`	Vertex or Route
Toggle Sign (+/-)	`T`	Route (or Edge for default)

Move Node	Left-Click Drag	Vertex
Route Edge	Left-Click Drag	Route

Developer Note: This tool is an authentic collaboration between human and AI, optimized for researchers and students working with signed planar graphs.